

# TDDD25: Distributed Systems Programming Project

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# Contact

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# Organisation

`https://www.ida.liu.se/~TDDD25/`

- 1 teaching session
- 9 lab sessions\*
- 1 + 5 labs
- 1 group, this year!

## Organisation (the same subject continued)

<https://www.ida.liu.se/~TDDD25/>

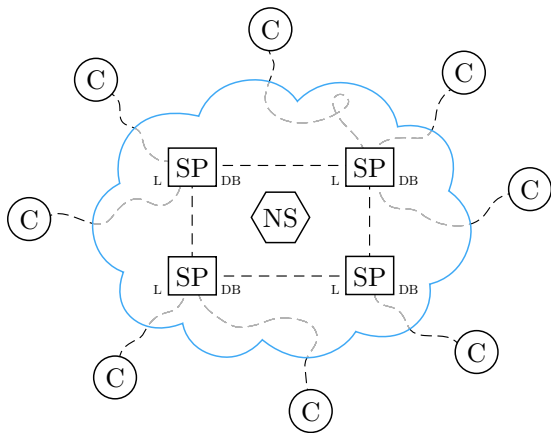
- 2 students per lab pair
- Registration deadline: January 22, 2026
- Completion deadline: March 11 – last lab session
- Rules!

<https://www.ida.liu.se/~TDDD25/labs/rules.en.shtml>

# Submission

- No written reports are needed (over and above the code, and anything requested from you in particular).
- Make your code available on GitLab.
- Add your teaching assistant as a Reporter *with access expiring 2026-12-31*.
- Demonstrate your solutions in class *and be prepared to discuss the concepts and conceptual questions in the labs*.
- **Attend your lab sessions.**

## Structure of the lab series



- C — client
- SP — server/peer
- NS — name service
- DB — database
- L — lock

# Source Code

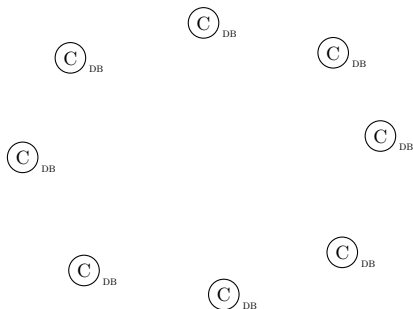
`https://gitlab.liu.se/LiU-TDDD25/labs/`

- doc/
- src/
  - lab0/
  - lab1/
  - lab2/
  - lab3/
  - lab4/
  - lab5/
  - modules/
    - Common/
    - Server/

# Implementation

- Multi-threaded object-oriented code in Python 3
- Communication via objects serialized in JSON
- Data transfer through TCP sockets
- `https://www.ida.liu.se/~TDDD25/labs/index.en.shtml#guidoscorner`

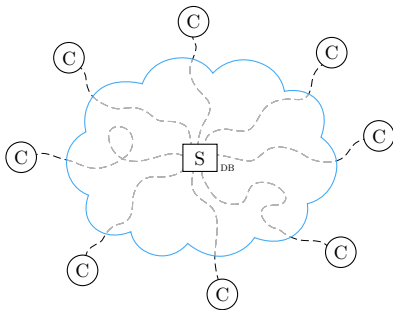
## Lab 0: Standalone Database



- Local database for each client (C)
- **TODO**: complete the implementation of the read and write operations of the database (DB)

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab0.pdf>

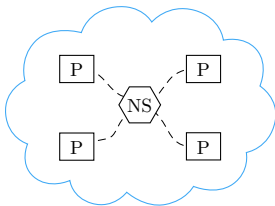
# Lab 1: Client-Server Database



- Centralized database (DB)
- **TODO:** complete the implementation of the client (C) and server (S) communication mechanism

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab1.pdf>

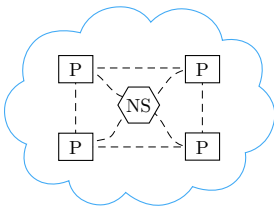
## Lab 2: Object Request Broker



- Name service (NS) and object request broker (ORB)
- Abstract away the communication part of the functionality
- **TODO**: complete the implementation of the ORB

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab2.pdf>

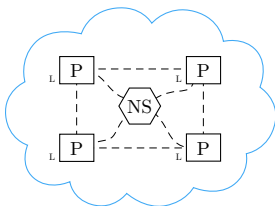
## Lab 3: Peer-to-Peer Communication



- Smart mechanism for keeping track of peers (P)
- **TODO**: complete the functionality dealing with the peers joining or leaving the system

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab3.pdf>

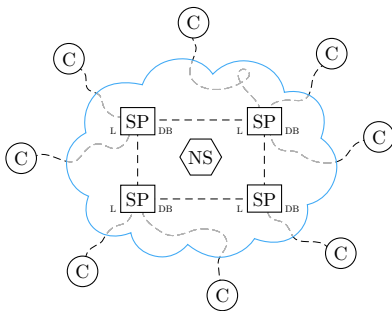
## Lab 4: Distributed Locks



- Distributed mutual exclusion to control concurrent operations
- **TODO**: complete the implementation of the second Ricart–Agrawala algorithm (L)

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab4.pdf>

## Lab 5: Client-Server Database with Replicas



- Everything together
- **TODO:** complete the implementation of the server/peer (SP) using all the previously developed components

<https://gitlab.liu.se/LiU-TDDD25/labs/raw/master/doc/lab5.pdf>

## Where do we perform the labs?

- Typically lab room hosts.
- Labs should be demonstrated on LiU systems, regardless of where they were written.
- Lab 2-5 requires opening connections to ns-tddd25.edu.liu.se on odd ports, *and the reverse*. People have managed using tunnel settings, but I don't provide support<sup>1</sup>
- Working from home? The student VPN is IPv4-only and split-tunnel.
- ThinLinc or RDP to the rescue!

<https://liuonline.sharepoint.com/sites/student-under-studietiden/SitePages/en/Fjarrinloggning.aspx>

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<sup>1</sup>Writing your own nameservice isn't that hard, but not part of the lab series.

# Speed-dating

- Has everyone got a lab partner?
- ...with compatible prerequisites, schedules, ambitions...?

Good luck!